

Computational Verification of the Dual Laws of Market Dynamics

AI, Machine Learning, Reinforcement Learning, and
Strategic Stress Testing for Liquidity, Preferability, and
Geeconomic Market Power

Soumadeep Ghosh

Kolkata, India

Abstract

This paper develops a computational verification framework for the Dual Laws of Market Dynamics. The first law, the Natural Law of Market Expansion, states that liquidity provision expands markets when marginal liquidity reduces effective transaction costs more than it increases adverse-selection, inventory, congestion, or strategic risk. The second law, the Natural Law of Preferable Markets, states that capital flows toward venues with superior marginal risk-adjusted strategic returns. We transform these laws into falsifiable inequalities and propose a research architecture using econometrics, causal machine learning, graph learning, deep order-flow models, natural-language geopolitical risk indices, multi-agent simulation, and reinforcement learning. The paper contributes a unified framework connecting market microstructure, finance, economics, political science, international relations, geoeconomics, warfare economics, cyber conflict, AI, and statistical verification.

The paper ends with “The End”

Contents

1	Introduction	2
2	Notation	3
3	The Dual Laws as Testable Scientific Claims	3
3.1	The Natural Law of Market Expansion	3
3.2	The Natural Law of Preferable Markets	4
4	Causal Structure of the Laws	5
5	Empirical Hypotheses	5
6	Data Architecture	5
7	Econometric Verification	5
7.1	Panel Model for Market Expansion	5
7.2	Capital Allocation Model	6
7.3	Endogeneity and Identification	6
8	Causal Machine Learning	6
9	Machine Learning Components	7
10	Deep Learning for Toxicity Detection	7
11	Graph Learning and Network Externalities	7
12	Geopolitical and Warfare Risk Measurement	8
12.1	Geopolitical Risk Index	8
12.2	Warfare Economics	8
13	Reinforcement Learning Verification	9
13.1	MDP Formulation	9
13.2	RL Verification Criterion	9

14 Algorithmic Verification Pipeline	9
15 Multi-Agent Simulation	10
15.1 Simulation Test of Law I	10
15.2 Simulation Test of Law II	10
16 Bayesian Verification	11
17 TikZ Illustration of Verification Architecture	11
18 Stylized Numerical Illustration	12
19 Failure Modes and Boundary Conditions	12
20 Strategic Market Power	13
21 Integrated Verification Criteria	13
22 Conclusion	13

List of Figures

1 Causal mechanism behind the Natural Law of Market Expansion. Liquidity expands markets through cost reduction, but toxicity and strategic risk can weaken or reverse the effect.	5
2 Market as a strategic network. Network centrality, dependency, and chokepoints affect liquidity expansion and geoeconomic power.	8
3 End-to-end AI/ML/RL verification architecture for the Dual Laws of Market Dynamics.	11
4 Stylized market expansion. Liquidity initially expands participation, but risk amplification can flatten or reverse the expansion effect.	12

List of Tables

1 Core notation	3
2 Empirical hypotheses implied by the Dual Laws	5
3 Data required for computational verification	5
4 Machine learning methods for verifying the Dual Laws	7
5 Conflict regimes and expected market effects	8
6 Agent types in the verification simulator	10
7 Failure modes of the Dual Laws	12

1 Introduction

Liquidity provision is traditionally studied as a market microstructure function: market-makers stand ready to buy and sell, earn bid-ask spreads, absorb inventory risk, and reduce transaction costs. Yet in modern financial systems, liquidity provision is also a strategic capability. States, exchanges, market-makers, clearinghouses, payment networks, settlement rails, and digital platforms jointly determine who can trade, at what cost, under what constraints, and under whose infrastructure.

The Dual Laws of Market Dynamics provide a theoretical foundation for this broader interpretation. The first law concerns market expansion: liquidity reduces transaction costs, which expands participation, which reinforces liquidity through network externalities. The second law concerns market preferability: market-makers and strategic capital allocators move capital toward venues with superior risk-adjusted returns after accounting for toxicity, geopolitical risk, warfare risk, and institutional constraints.

This paper does not merely restate these laws. It asks a sharper question:

Can AI, machine learning, reinforcement learning, and data science verify, falsify, or refine the Dual Laws of Market Dynamics?

The answer is affirmative, provided the laws are expressed as measurable, falsifiable, and stress-testable claims.

2 Notation

Table 1: Core notation

Symbol	Meaning
M_t	Market size or active participation at time t
L_t	Liquidity provision at time t
C_t	Effective transaction cost
N_t	Number of potential market participants
s_t	Bid-ask spread
D_t	Market depth
σ_t	Volatility
I_t	Inventory exposure
A_t	Adverse-selection parameter
T_t	Order-flow toxicity
G_t	Geopolitical risk index
W_t	Warfare or conflict risk index
$\Phi_{j,t}$	Preferability index of market j at time t
$k_{j,t}$	Capital allocated to market j at time t
$\pi_{j,t}$	Profit from market j at time t
K_t	Total available capital
μ_t	Shadow price of capital

3 The Dual Laws as Testable Scientific Claims

3.1 The Natural Law of Market Expansion

The basic form of the Natural Law of Market Expansion is

$$L_t \uparrow \Rightarrow C_t \downarrow \Rightarrow M_t \uparrow. \quad (1)$$

In differential form, the law asserts

$$\frac{\partial M_t}{\partial L_t} > 0 \quad (2)$$

whenever

$$\frac{\partial C_t}{\partial L_t} < 0. \quad (3)$$

However, a stronger formulation must recognize that liquidity can also increase adverse selection, predatory trading, congestion, leverage, or systemic fragility. Let R_t denote the composite risk generated by additional liquidity:

$$R_t = R(T_t, I_t, \sigma_t, G_t, W_t). \quad (4)$$

Then effective participation depends on both costs and risks:

$$M_t = F(C_t, R_t, N_t). \quad (5)$$

Proposition 1 (Risk-Adjusted Market Expansion). *Liquidity provision expands market size if and only if the marginal cost reduction dominates the marginal risk increase:*

$$-\frac{\partial F}{\partial C_t} \frac{\partial C_t}{\partial L_t} > -\frac{\partial F}{\partial R_t} \frac{\partial R_t}{\partial L_t}. \quad (6)$$

Equivalently,

$$\frac{\partial M_t}{\partial L_t} > 0. \quad (7)$$

Proof. By the chain rule,

$$\frac{\partial M_t}{\partial L_t} = \frac{\partial F}{\partial C_t} \frac{\partial C_t}{\partial L_t} + \frac{\partial F}{\partial R_t} \frac{\partial R_t}{\partial L_t}. \quad (8)$$

Participation is decreasing in costs and risk, so

$$\frac{\partial F}{\partial C_t} < 0, \quad \frac{\partial F}{\partial R_t} < 0. \quad (9)$$

Liquidity normally reduces transaction costs, so

$$\frac{\partial C_t}{\partial L_t} < 0. \quad (10)$$

The liquidity effect is positive when the positive cost-reduction channel exceeds the negative risk-amplification channel:

$$\frac{\partial F}{\partial C_t} \frac{\partial C_t}{\partial L_t} + \frac{\partial F}{\partial R_t} \frac{\partial R_t}{\partial L_t} > 0. \quad (11)$$

Rearranging gives the desired condition. $\square$

3.2 The Natural Law of Preferable Markets

The second law concerns capital allocation across heterogeneous markets. A market-maker allocates capital toward venues with superior marginal risk-adjusted returns.

Let the certainty equivalent of market j be

$$\text{CE}_{j,t} = \mathbb{E}[\pi_{j,t}] - \frac{\gamma}{2} \text{Var}(\pi_{j,t}) - \eta T_{j,t} - \delta G_{j,t} - \zeta W_{j,t}, \quad (12)$$

where γ measures risk aversion, η toxicity aversion, δ geopolitical-risk aversion, and ζ warfare-risk aversion.

The preferability index is

$$\Phi_{j,t} = \frac{\mathbb{E}[\pi_{j,t}]}{\sqrt{\text{Var}(\pi_{j,t})}} - \eta T_{j,t} - \delta G_{j,t} - \zeta W_{j,t}. \quad (13)$$

Theorem 1 (Natural Law of Preferable Markets). *Given capital constraint*

$$\sum_{j=1}^J k_{j,t} \leq K_t, \quad (14)$$

optimal capital allocation satisfies

$$k_{j,t}^* > 0 \iff \frac{\partial \text{CE}_{j,t}}{\partial k_{j,t}} \geq \mu_t, \quad (15)$$

where μ_t is the shadow price of capital. Markets with marginal certainty-equivalent return below μ_t receive zero capital.

Proof. The market-maker solves

$$\max_{\{k_{j,t}\}_{j=1}^J} \sum_{j=1}^J \text{CE}_{j,t}(k_{j,t}) \quad (16)$$

subject to

$$\sum_{j=1}^J k_{j,t} \leq K_t, \quad k_{j,t} \geq 0. \quad (17)$$

The Lagrangian is

$$\mathcal{L} = \sum_{j=1}^J \text{CE}_{j,t}(k_{j,t}) - \mu_t \left(\sum_{j=1}^J k_{j,t} - K_t \right) + \sum_{j=1}^J \lambda_{j,t} k_{j,t}. \quad (18)$$

The Karush–Kuhn–Tucker conditions imply

$$\frac{\partial \text{CE}_{j,t}}{\partial k_{j,t}} - \mu_t + \lambda_{j,t} = 0, \quad (19)$$

with complementary slackness

$$\lambda_{j,t} k_{j,t} = 0. \quad (20)$$

If $k_{j,t}^* > 0$, then $\lambda_{j,t} = 0$, hence

$$\frac{\partial \text{CE}_{j,t}}{\partial k_{j,t}} = \mu_t. \quad (21)$$

If $k_{j,t}^* = 0$, then $\lambda_{j,t} \geq 0$, so

$$\frac{\partial \text{CE}_{j,t}}{\partial k_{j,t}} \leq \mu_t. \quad (22)$$

Thus active markets are those whose marginal certainty-equivalent return is at least the opportunity cost of capital. $\square$

4 Causal Structure of the Laws

The verification problem can be represented as a causal graph.

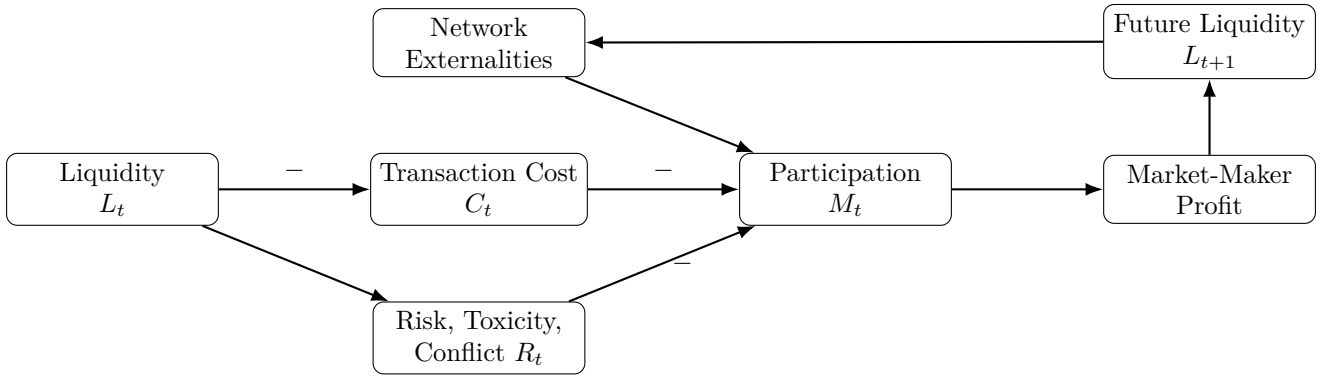


Figure 1: Causal mechanism behind the Natural Law of Market Expansion. Liquidity expands markets through cost reduction, but toxicity and strategic risk can weaken or reverse the effect.

5 Empirical Hypotheses

The laws imply falsifiable hypotheses.

Table 2: Empirical hypotheses implied by the Dual Laws

Hypothesis	Mathematical claim	Expected sign
H1: Liquidity lowers costs	$\partial C_t / \partial L_t$	Negative
H2: Costs reduce participation	$\partial M_t / \partial C_t$	Negative
H3: Liquidity expands markets	$\partial M_t / \partial L_t$	Positive
H4: Toxicity weakens expansion	$\partial^2 M_t / \partial L_t \partial T_t$	Negative
H5: Geopolitical risk lowers preferability	$\partial \Phi_{j,t} / \partial G_{j,t}$	Negative
H6: Warfare risk lowers allocation	$\partial k_{j,t} / \partial W_{j,t}$	Negative
H7: Capital follows preferability	$\partial k_{j,t} / \partial \Phi_{j,t}$	Positive
H8: Network effects create thresholds	M_t nonlinear in L_t	Convex near L^*

6 Data Architecture

Table 3: Data required for computational verification

Data class	Variables	Purpose
Limit order book	Depth, spread, imbalance	Estimate liquidity and costs
Trades	Price, size, direction, volume	Estimate market size and flow toxicity
Volatility data	Realized volatility, jumps	Control for market risk
Venue data	Fees, latency, rules, outages	Explain market preferability
Capital data	Inventory, allocation, limits	Test allocation law
News data	Sanctions, conflict, diplomacy	Construct geopolitical risk
Cyber data	Attacks, outages, breaches	Construct infrastructure risk
Macro data	Rates, inflation, FX, growth	Control for economic regimes
Network data	Clearing, settlement, dependencies	Estimate strategic market power

7 Econometric Verification

7.1 Panel Model for Market Expansion

A baseline panel specification is

$$\Delta \ln M_{i,t} = \alpha_i + \tau_t + \beta_1 \Delta L_{i,t-1} + \beta_2 \Delta C_{i,t} + \beta_3 T_{i,t} + \beta_4 G_{i,t} + \boldsymbol{\theta}' \mathbf{X}_{i,t} + \varepsilon_{i,t}, \quad (23)$$

where i indexes markets or venues, α_i are market fixed effects, and τ_t are time fixed effects. The first law predicts

$$\beta_1 > 0, \quad \beta_2 < 0. \quad (24)$$

7.2 Capital Allocation Model

For the second law,

$$k_{j,t+1} = \alpha_j + \tau_t + \rho k_{j,t} + \theta \Phi_{j,t} + \boldsymbol{\lambda}' \mathbf{Z}_{j,t} + u_{j,t}. \quad (25)$$

The Natural Law of Preferable Markets predicts

$$\theta > 0. \quad (26)$$

7.3 Endogeneity and Identification

Liquidity, participation, spreads, volatility, and capital allocation are jointly determined. Therefore ordinary least squares may be biased. Identification strategies include:

1. instrumental variables using fee changes, tick-size changes, or exogenous venue outages;
2. difference-in-differences around regulatory or infrastructure shocks;
3. panel vector autoregression for dynamic feedback;
4. structural estimation of market-maker objective functions;
5. causal machine learning for heterogeneous treatment effects.

8 Causal Machine Learning

Let treatment be liquidity:

$$T_t = L_t. \quad (27)$$

Let outcome be market expansion:

$$Y_t = \Delta \ln M_t. \quad (28)$$

Let controls be

$$\mathbf{X}_t = (\sigma_t, D_t, s_t, T_t, G_t, W_t, \text{fees}_t, \text{macro}_t). \quad (29)$$

A partially linear causal model is

$$Y_t = \theta T_t + g(\mathbf{X}_t) + \varepsilon_t, \quad (30)$$

$$T_t = m(\mathbf{X}_t) + v_t. \quad (31)$$

Double machine learning estimates nuisance functions g and m flexibly, then estimates θ using orthogonalized residuals:

$$\tilde{Y}_t = Y_t - \hat{g}(\mathbf{X}_t), \quad \tilde{T}_t = T_t - \hat{m}(\mathbf{X}_t). \quad (32)$$

The causal effect is estimated by

$$\hat{\theta} = \frac{\sum_t \tilde{T}_t \tilde{Y}_t}{\sum_t \tilde{T}_t^2}. \quad (33)$$

The first law is supported if

$$\hat{\theta} > 0 \quad (34)$$

across normal regimes and if the effect weakens or reverses in high-toxicity or warfare-stress regimes.

9 Machine Learning Components

Table 4: Machine learning methods for verifying the Dual Laws

Method	Target	Role in verification
Random forests	C_t, M_t	Nonlinear benchmark
Gradient boosting	Costs, spreads, participation	High-performance tabular prediction
Neural networks	Order-flow dynamics	Flexible nonlinear approximation
LSTM/Transformer	Sequential order flow	Forecast toxicity and liquidity decay
Causal forests	Treatment effects	Estimate heterogeneous liquidity effects
Graph neural networks	Network externalities	Detect critical mass and centrality effects
Bayesian models	Parameter uncertainty	Posterior law verification
Reinforcement learning	Capital allocation	Test preferability through optimal policy
Multi-agent RL	Strategic competition	Model adversarial market-making

10 Deep Learning for Toxicity Detection

Order-flow toxicity is central because informed trading can make liquidity provision unprofitable and destabilizing. Define volume-synchronized probability of informed trading:

$$\text{VPIN}_t = \frac{\sum_{i=1}^{\tau} |V_i^B - V_i^S|}{\tau V}, \quad (35)$$

where V_i^B and V_i^S are buy and sell volumes in volume bucket i .
Let the feature vector be

$$\mathbf{x}_t = (\text{VPIN}_t, \sigma_t, \text{OFI}_t, s_t, D_t, \Delta p_t, Q_t). \quad (36)$$

A neural toxicity classifier is

$$\hat{T}_t = \sigma(W_2 \text{ReLU}(W_1 \mathbf{x}_t + \mathbf{b}_1) + b_2). \quad (37)$$

The conditional expansion law becomes

$$\frac{\partial M_t}{\partial L_t} > 0 \quad \text{when} \quad \hat{T}_t < \tau_T, \quad (38)$$

but

$$\frac{\partial M_t}{\partial L_t} \leq 0 \quad \text{may occur when} \quad \hat{T}_t \geq \tau_T. \quad (39)$$

11 Graph Learning and Network Externalities

Markets are networks. Participants, venues, clearinghouses, custodians, market-makers, states, banks, payment systems, and information platforms form a strategic graph.

Let

$$\mathcal{G}_t = (\mathcal{V}_t, \mathcal{E}_t) \quad (40)$$

be the market network. A graph neural network estimates

$$\hat{M}_{t+1} = \text{GNN}(\mathcal{G}_t, L_t, C_t, R_t). \quad (41)$$

A network threshold exists if there is L^* such that

$$\frac{\partial^2 M_t}{\partial L_t^2} > 0 \quad \text{near} \quad L^*. \quad (42)$$

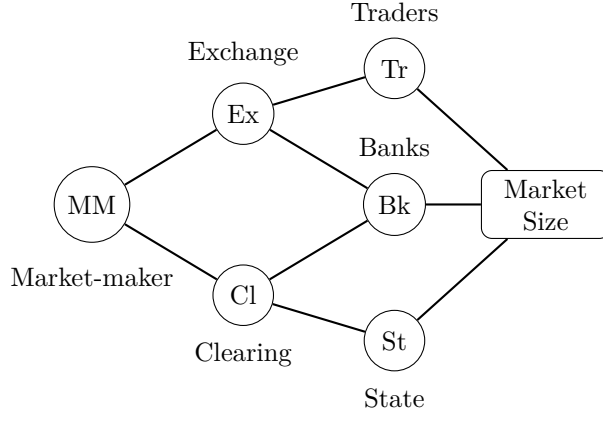


Figure 2: Market as a strategic network. Network centrality, dependency, and chokepoints affect liquidity expansion and geoeconomic power.

12 Geopolitical and Warfare Risk Measurement

12.1 Geopolitical Risk Index

Let $d_{i,t}$ denote document i at time t , including news, official statements, sanctions releases, diplomatic communications, and cyber warnings. A transformer-based geopolitical risk index is

$$G_t = \frac{1}{N_t} \sum_{i=1}^{N_t} \text{Sentiment}(d_{i,t}) \cdot \text{Relevance}(d_{i,t}) \cdot \text{Severity}(d_{i,t}). \quad (43)$$

The geopolitical channel of the second law predicts

$$G_{j,t} \uparrow \Rightarrow \Phi_{j,t} \downarrow \Rightarrow k_{j,t+1} \downarrow. \quad (44)$$

12.2 Warfare Economics

Conflict changes liquidity through infrastructure risk, sanctions, payment disruption, cyberattack, capital controls, and physical destruction.

Table 5: Conflict regimes and expected market effects

Regime	Liquidity effect	Spread effect	Strategic response
Trade war	Moderate decline	2–3× normal	Venue diversification
Sanctions	Asymmetric decline	Venue-specific widening	Regulatory arbitrage
Cyberattack	Severe temporary decline	5–10× normal	Redundant systems
Naval blockade	Commodity-specific decline	High basis volatility	Reroute supply chains
Kinetic warfare	Critical collapse	Extreme widening	Cease or emergency operation
Financial war	Chokepoint pressure	Fragmented liquidity	Shift settlement networks

Let warfare risk enter adverse selection as

$$A_t = A_0 \exp(\psi W_t), \quad (45)$$

where $\psi > 0$. Then expected profit becomes

$$\mathbb{E}[\pi_{j,t}] = s_{j,t} Q_{j,t} - \lambda_{j,t} \sigma_{j,t}^2 I_{j,t}^2 - A_0 e^{\psi W_{j,t}} Q_{j,t}^{\text{toxic}} - K_{j,t}. \quad (46)$$

Thus

$$\frac{\partial \mathbb{E}[\pi_{j,t}]}{\partial W_{j,t}} = -\psi A_0 e^{\psi W_{j,t}} Q_{j,t}^{\text{toxic}} < 0. \quad (47)$$

13 Reinforcement Learning Verification

13.1 MDP Formulation

Capital allocation is naturally sequential. Define the state as

$$s_t = (\Phi_{1,t}, \dots, \Phi_{J,t}, K_t, I_t, T_t, G_t, W_t, \sigma_t). \quad (48)$$

The action is

$$a_t = (k_{1,t}, \dots, k_{J,t}), \quad (49)$$

with

$$\sum_{j=1}^J k_{j,t} \leq K_t. \quad (50)$$

The reward is

$$r_t = \sum_{j=1}^J \pi_{j,t} - \lambda_1 \text{InventoryRisk}_t - \lambda_2 \text{Drawdown}_t - \lambda_3 G_t - \lambda_4 W_t. \quad (51)$$

The Bellman equation is

$$Q^*(s, a) = \mathbb{E} \left[r_t + \beta \max_{a'} Q^*(s_{t+1}, a') \mid s_t = s, a_t = a \right]. \quad (52)$$

13.2 RL Verification Criterion

The second law is computationally supported if the learned policy π_θ satisfies

$$\frac{\partial k_{j,t}^{\pi_\theta}}{\partial \Phi_{j,t}} > 0 \quad (53)$$

and

$$k_{j,t}^{\pi_\theta} \rightarrow 0 \quad \text{when} \quad \Phi_{j,t} < \bar{\Phi}_t. \quad (54)$$

Proposition 2 (RL Recovery of Preferable Markets). *Assume rewards are increasing in $\Phi_{j,t}$ and decreasing in toxicity, geopolitical risk, and warfare risk. If the reinforcement learning algorithm converges to an optimal policy, then the learned allocation is monotone in marginal market preferability.*

Proof. If rewards are increasing in $\Phi_{j,t}$, then for two otherwise identical markets j and ℓ with $\Phi_{j,t} > \Phi_{\ell,t}$, allocating an additional unit of capital to j yields a weakly higher expected immediate reward. If transition dynamics preserve monotonicity, the continuation value is also weakly higher. Therefore the Bellman optimality operator preserves the preference ordering. At convergence, the optimal policy allocates weakly more capital to the market with higher marginal preferability, subject to risk limits and capacity constraints. $\square$

14 Algorithmic Verification Pipeline

Algorithm 1 Computational Verification of the Dual Laws

Require: Order book data, trade data, venue data, macro data, geopolitical text, cyber-risk data

Ensure: Verification statistics for market expansion and preferable-market allocation

- 1: Estimate liquidity L_t , transaction cost C_t , market size M_t
 - 2: Estimate toxicity $\hat{T}_t$ using order-flow features
 - 3: Estimate geopolitical risk $\hat{G}_t$ using NLP
 - 4: Estimate warfare and infrastructure risk $\hat{W}_t$
 - 5: Fit cost model $\hat{C}_t = f(L_t, \mathbf{X}_t)$
 - 6: Fit participation model $\hat{M}_t = g(C_t, L_t, R_t, \mathbf{X}_t)$
 - 7: Use causal ML to estimate $\partial \hat{M} / \partial L$
 - 8: Compute preferability indices $\hat{\Phi}_{j,t}$
 - 9: Train RL allocator with state s_t and action a_t
 - 10: Test whether $\partial \hat{k}_{j,t} / \partial \hat{\Phi}_{j,t} > 0$
 - 11: Stress-test under sanctions, cyberattacks, volatility shocks, and war regimes
 - 12: Report normal-regime support, crisis-regime failure, and boundary conditions
-

Algorithm 2 Reinforcement Learning Capital Allocator

Require: Markets $j = 1, \dots, J$, capital K_t , risk limits, reward parameters

- 1: Initialize policy $\pi_\theta(a_t|s_t)$
 - 2: **for** episode = 1 to N **do**
 - 3: Observe state $s_t = (\Phi_{1,t}, \dots, \Phi_{J,t}, K_t, T_t, G_t, W_t)$
 - 4: Choose allocation $a_t = (k_{1,t}, \dots, k_{J,t}) \sim \pi_\theta$
 - 5: Enforce $\sum_j k_{j,t} \leq K_t$
 - 6: Execute quotes and hedges in simulated or historical environment
 - 7: Observe profits $\pi_{j,t}$, toxicity, drawdown, and risk penalties
 - 8: Compute reward r_t
 - 9: Update policy parameters θ
 - 10: **end for**
 - 11: Return learned policy π_θ^*
-

15 Multi-Agent Simulation

A multi-agent market simulator can verify whether the laws emerge endogenously.

Table 6: Agent types in the verification simulator

Agent	Objective	Strategic role
Noise trader	Liquidity demand	Generates baseline volume
Informed trader	Trade on private signal	Creates adverse selection
Market-maker	Earn spreads, manage inventory	Provides liquidity
Exchange	Maximize volume and fees	Sets rules and fees
Regulator	Control systemic risk	Imposes constraints
State actor	Strategic leverage	Uses sanctions or access control
Cyber actor	Disrupt infrastructure	Creates operational risk
Arbitrageur	Exploit price differences	Links venues

15.1 Simulation Test of Law I

Create two otherwise identical markets A and B :

$$L_A > L_B. \tag{55}$$

The first law predicts

$$C_A < C_B \tag{56}$$

and

$$M_A > M_B. \tag{57}$$

15.2 Simulation Test of Law II

Create heterogeneous venues with different spreads, depth, toxicity, volatility, sanctions risk, and cyber risk. The second law predicts that RL market-makers allocate capital such that

$$\Phi_A > \Phi_B \quad \Rightarrow \quad k_A^* \geq k_B^*, \tag{58}$$

subject to capacity and diversification constraints.

Algorithm 3 Multi-Agent Stress Test

Require: Number of markets J , agent populations, conflict scenarios

- 1: **for** scenario in normal, toxic flow, sanctions, cyberattack, war **do**
- 2: Initialize markets with heterogeneous liquidity and risk parameters
- 3: Initialize agents: noise traders, informed traders, market-makers, exchanges, state actors
- 4: **for** $t = 1$ to T **do**
- 5: Agents submit orders according to their strategies
- 6: Market-makers update quotes and inventories
- 7: Exchanges clear trades and update order books
- 8: State or cyber actors impose scenario-specific shocks
- 9: Record $L_t, C_t, M_t, \Phi_t, k_t, \pi_t$
- 10: **end for**
- 11: Estimate law-verification inequalities
- 12: **end for**
- 13: Compare support across regimes

16 Bayesian Verification

A Bayesian framework quantifies uncertainty in the laws.

Let

$$\theta_i = \frac{\partial M_i}{\partial L_i} \quad (59)$$

be the market-specific expansion effect. A hierarchical model is

$$\theta_i \sim \mathcal{N}(\mu_\theta, \sigma_\theta^2), \quad (60)$$

$$\mu_\theta \sim \mathcal{N}(0, 10), \quad (61)$$

$$\sigma_\theta \sim \text{HalfCauchy}(0, 5). \quad (62)$$

The posterior probability that the first law holds is

$$\Pr(\theta_i > 0 \mid \mathcal{D}). \quad (63)$$

Similarly, for the second law let

$$\omega_j = \frac{\partial k_j}{\partial \Phi_j}. \quad (64)$$

The second law is supported when

$$\Pr(\omega_j > 0 \mid \mathcal{D}) \quad (65)$$

is high across markets and regimes.

17 TikZ Illustration of Verification Architecture

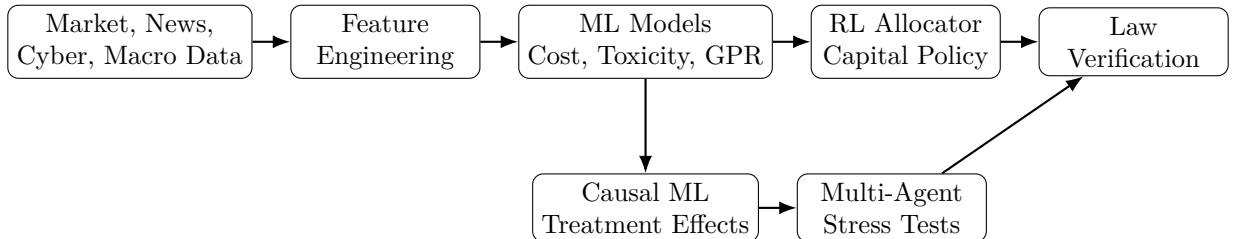


Figure 3: End-to-end AI/ML/RL verification architecture for the Dual Laws of Market Dynamics.

18 Stylized Numerical Illustration

Suppose

$$C(L) = c_0 e^{-aL}, \quad R(L) = r_0 + bL^2, \quad (66)$$

and

$$M(L) = N [1 - F(C(L) + R(L))]. \quad (67)$$

Then

$$\frac{dM}{dL} = -N f(C + R) (C'(L) + R'(L)). \quad (68)$$

Since $f > 0$, expansion requires

$$C'(L) + R'(L) < 0. \quad (69)$$

Because

$$C'(L) = -ac_0 e^{-aL}, \quad R'(L) = 2bL, \quad (70)$$

the expansion condition is

$$ac_0 e^{-aL} > 2bL. \quad (71)$$

Thus liquidity expands the market at low and moderate liquidity levels but may fail at extreme liquidity levels if risk grows faster than cost reduction.

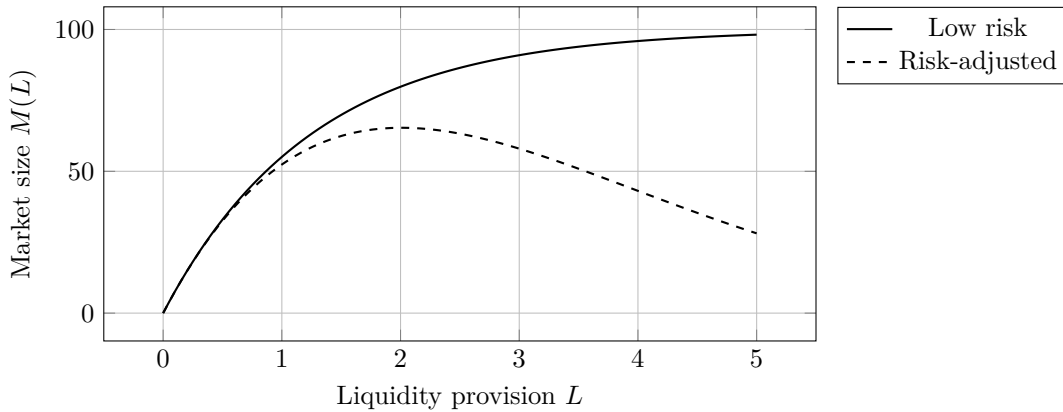


Figure 4: Stylized market expansion. Liquidity initially expands participation, but risk amplification can flatten or reverse the expansion effect.

19 Failure Modes and Boundary Conditions

The laws are not unconditional. They may fail under identifiable conditions.

Table 7: Failure modes of the Dual Laws

Failure mode	Law affected	Mechanism
Toxic order flow	Expansion and preferability	Adverse selection overwhelms spread revenue
Flash crash	Expansion	Liquidity vanishes during stress
Cyberattack	Expansion	Infrastructure failure raises execution risk
Sanctions	Preferability	Access risk lowers expected returns
Capital controls	Preferability	Exit constraints reduce allocation attractiveness
War	Both	Physical, financial, and informational disruption
Predatory liquidity	Expansion	Apparent liquidity becomes destabilizing
Fragmented venues	Both	Network benefits weaken across disconnected markets

20 Strategic Market Power

A state or institutional bloc possesses strategic market power when it controls liquidity, infrastructure, standards, settlement networks, or critical nodes.

Define strategic market power in asset class a as

$$\text{SMP}_{a,t} = \frac{L_{a,t}^{\text{domestic}}}{L_{a,t}^{\text{global}}} (1 + \chi \text{Centrality}_{a,t}). \quad (72)$$

If state A imposes sanctions on state B , the cost to B is

$$\text{Cost}_{B,t} = \theta \cdot \text{SMP}_{A,t} \cdot \text{Dependency}_{B \rightarrow A,t} \cdot \mathbb{I}\{\text{sanctions}\}. \quad (73)$$

This links market microstructure to geoeconomics and international relations: control over market infrastructure becomes a coercive instrument.

21 Integrated Verification Criteria

The Dual Laws are computationally supported when the following inequalities hold across datasets, simulations, and regimes:

$$\widehat{\frac{\partial C}{\partial L}} < 0, \quad (74)$$

$$\widehat{\frac{\partial M}{\partial C}} < 0, \quad (75)$$

$$\widehat{\frac{\partial M}{\partial L}} > 0, \quad (76)$$

$$\widehat{\frac{\partial k_j}{\partial \Phi_j}} > 0, \quad (77)$$

$$\widehat{\frac{\partial k_j}{\partial G_j}} < 0, \quad (78)$$

$$\widehat{\frac{\partial k_j}{\partial W_j}} < 0. \quad (79)$$

The strongest confirmation arises when these inequalities hold out of sample, across venues, asset classes, regulatory systems, geopolitical regimes, and simulated adversarial environments.

22 Conclusion

The Dual Laws of Market Dynamics can be transformed from theoretical propositions into a computational scientific program. The Natural Law of Market Expansion becomes a testable causal claim about liquidity, costs, risk, and participation. The Natural Law of Preferable Markets becomes a testable optimization claim about capital allocation under risk, toxicity, geopolitical stress, and warfare conditions.

AI and machine learning provide the measurement layer. Causal ML provides identification. Graph learning captures network externalities. NLP quantifies geopolitical risk. Deep learning detects toxicity. Reinforcement learning tests whether optimal agents discover the laws through interaction. Multi-agent simulation reveals boundary conditions and failure regimes.

The resulting framework treats market-making as both an economic function and a strategic capability. In normal markets, liquidity expands participation. In preferable markets, capital follows risk-adjusted opportunity. In conflict regimes, liquidity becomes infrastructure, capital allocation becomes strategy, and market control becomes geoeconomic power.

References

- [1] Acemoglu, D., Chernozhukov, V., Werning, I., and Whinston, M. D. (2015). Optimal redistribution in the presence of signaling. *American Economic Review*, 105(1), 1–39.
- [2] Blackwill, R. D., and Harris, J. M. (2016). *War by Other Means: Geoeconomics and Statecraft*. Harvard University Press.
- [3] Easley, D., Lopez de Prado, M., and O’Hara, M. (2012). Flow toxicity and liquidity in a high-frequency world. *Review of Financial Studies*, 25(5), 1457–1493.
- [4] Farrell, H., and Newman, A. L. (2019). Weaponized interdependence: How global economic networks shape state coercion. *International Security*, 44(1), 42–79.
- [5] Glosten, L. R., and Milgrom, P. R. (1985). Bid, ask and transaction prices in a specialist market with heterogeneously informed traders. *Journal of Financial Economics*, 14(1), 71–100.
- [6] Ghosh, S. (2026). The Dual Laws of Market Dynamics: An Interdisciplinary Framework for Liquidity Provision in Geoeconomic Competition. Working paper.
- [7] Kyle, A. S. (1985). Continuous auctions and insider trading. *Econometrica*, 53(6), 1315–1335.
- [8] Lopez de Prado, M. (2018). *Advances in Financial Machine Learning*. Wiley.
- [9] O’Hara, M. (1997). *Market Microstructure Theory*. Blackwell.
- [10] Paszke, A., Gross, S., Massa, F., et al. (2019). PyTorch: An imperative style, high-performance deep learning library. *Advances in Neural Information Processing Systems*, 32.
- [11] Sutton, R. S., and Barto, A. G. (2018). *Reinforcement Learning: An Introduction*. MIT Press.
- [12] Vaswani, A., Shazeer, N., Parmar, N., et al. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, 30.

Glossary

Adverse Selection

The risk that a market-maker trades against better-informed counterparties.

AI Verification

The use of artificial intelligence methods to test whether theoretical market laws hold in empirical or simulated environments.

Capital Allocation

The distribution of financial resources across markets, venues, or assets.

Causal Machine Learning

A class of methods that estimates treatment effects while flexibly controlling for high-dimensional confounders.

Certainty Equivalent

A risk-adjusted value of uncertain profit, usually equal to expected profit minus a risk penalty.

Cyber Risk

The risk of market disruption from hacking, outages, data manipulation, or infrastructure compromise.

Geoeconomics

The use of economic instruments and infrastructure for geopolitical objectives.

Geopolitical Risk Index

A quantitative measure of conflict, sanctions, diplomatic tension, or political instability.

Graph Neural Network

A machine learning model designed to learn from network-structured data.

Liquidity Provision

The act of standing ready to buy and sell, thereby reducing transaction costs and facilitating trade.

Market Expansion

Growth in market participation, volume, depth, or economic reach.

Market Preferability

The attractiveness of a market after adjusting for expected returns, volatility, toxicity, geopolitical risk, and warfare risk.

Market Microstructure

The study of how trading rules, order books, dealers, and institutions affect prices and liquidity.

Multi-Agent Simulation

A computational environment in which heterogeneous agents interact strategically.

Network Externality

A condition where the value of a market rises with the number or centrality of participants.

Order-Flow Toxicity

The degree to which order flow comes from informed or strategically adverse traders.

Reinforcement Learning

A machine learning framework in which agents learn policies by interacting with an environment and maximizing cumulative reward.

Strategic Market Power

The ability to influence economic outcomes through control over liquidity, infrastructure, settlement, or network centrality.

Transaction Cost

The total cost of trading, including spreads, fees, market impact, delay, and adverse selection.

VPIN

Volume-Synchronized Probability of Informed Trading, a measure of order-flow imbalance and toxicity.

Warfare Economics

The study of how conflict, coercion, destruction, logistics, and state strategy affect economic systems.

The End